🔹 MLOps & Azure ML – Practical MCQs

1.

A data science team trained a model locally and wants to ensure reproducibility in production. What is the **best first step**?
A. Deploy the model directly
B. Save the model in a shared folder
C. Register the model with versioning
D. Convert model to ONNX

Answer: C

2.

Your model performs well initially but degrades after 3 months due to changing user behavior. What is this phenomenon?
A. Overfitting
B. Concept drift
C. Data leakage
D. Underfitting

Answer:B

3.

In Azure Machine Learning, what is the **primary purpose of a workspace**?
A. Store datasets only
B. Manage compute clusters
C. Central hub for all ML assets
D. Run notebooks

Answer:C

4.

A pipeline retrains a model whenever new data arrives in Blob storage. What type of pipeline trigger is this?
A. Manual trigger
B. Schedule-based trigger
C. Event-based trigger
D. Batch trigger

Answer:C

5.

Which component ensures that the **same environment (libraries, dependencies)** is used during training and deployment?
A. Dataset version
B. Compute instance
C. Environment configuration
D. Endpoint

Answer:C

6.

A team wants to compare multiple model versions and track metrics like accuracy and F1-score. What should they use?
A. Azure Monitor
B. Experiment tracking
C. Data factory
D. Key vault

Answer:B

7.

Which practice helps rollback to a previous working model in case of failure?
A. Logging
B. Version control of models
C. Data labeling
D. Feature scaling

Answer:B

8.

Your deployed endpoint suddenly starts returning incorrect predictions due to schema mismatch in input data. What is the issue?
A. Concept drift
B. Data drift
C. Training drift
D. Model bias

Answer:B

9.

In an enterprise setup, why is CI/CD integrated into MLOps pipelines?
A. To reduce model size
B. To automate testing and deployment
C. To improve model accuracy
D. To visualize data

Answer:B

10.

Which Azure ML feature allows you to **deploy a model as a REST API**?
A. Dataset
B. Pipeline
C. Endpoint
D. Compute cluster

Answer:C

11.

A model needs to be retrained weekly regardless of new data availability. Which approach is best?
A. Event trigger

B. Manual execution
C. Scheduled pipeline
D. Real-time inference

Answer:C

12.

Which metric would you monitor to detect **data drift** in production?
A. Accuracy
B. Precision
C. Input feature distribution change
D. Loss function

Answer:C

13.

In AIOps, what is the main goal?
A. Improve model training speed
B. Automate IT operations using AI
C. Reduce dataset size
D. Build dashboards

Answer:B

14.

A sudden spike in system errors is automatically detected and resolved using ML-based anomaly detection. This is an example of:
A. DevOps
B. MLOps
C. AIOps
D. DataOps

Answer:C

15.

Which Azure service is commonly integrated with Azure ML for logging and monitoring?
A. Azure DevOps
B. Azure Monitor
C. Azure Synapse
D. Azure Storage

Answer:B

16.

You want to ensure that only approved models are deployed to production. What should you implement?
A. Data pipeline
B. Model approval workflow
C. Data cleaning
D. Feature engineering

Answer:B

17.

A team wants to retrain models automatically when performance drops below a threshold. What is this called?
A. Continuous training
B. Batch scoring
C. Static deployment
D. Manual retraining

Answer:A

18.

Which artifact is stored when you register a model in Azure ML?
A. Only code
B. Only dataset
C. Model file with metadata
D. Logs only

Answer:C

19.

In production, a model is receiving data from a completely new region with different patterns. What should be done?
A. Ignore the data
B. Increase learning rate
C. Retrain model with new data
D. Delete endpoint

Answer:C

20.

Which component is responsible for orchestrating end-to-end ML workflows in Azure ML?
A. Notebook
B. Pipeline
C. Endpoint
D. Dataset

Answer:B